

Seyed Mahmood Shantiaeezade

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Summary

Robotics Ph.D. student at Oregon State University specializing in precision motion and vibration control for mechatronic systems. Experienced in system identification, dynamic modeling, and controller development, with advanced proficiency in MATLAB and Simulink and hands-on laboratory experience validating models and control performance on voice-coil actuators and machine-tool feed drives. Contributed to Team OSU's first-place finish in the 2025 ASPE Student Challenge.

Education

Ph.D. in Robotics

Sep. 2025 - Expected Sep. 2030

Oregon State University, Corvallis, OR | GPA: 3.88 / 4.00

M.Sc. in Mechanical Engineering

Sep. 2023 - Sep. 2025

Oregon State University, Corvallis, OR | GPA: 3.92 / 4.00

B.Sc. in Mechanical Engineering

Sep. 2018 - Jul. 2023

University of Tehran, Tehran | GPA: 3.72 / 4.00

Research and Engineering Experience

Graduate Research Assistant | Sep. 2023 - Present

Precision Manufacturing & Automation Laboratory

- Developed and experimentally validated spindle-synchronized X/Y feed motion to suppress milling vibration; MATLAB models and 5 kHz force measurements showed approximately **2-3x greater stable cutting depth** under the tested conditions.
- Developed and tested an **adaptive feedforward algorithm** that adjusts motion amplitude and phase from measured cutting forces, reducing unstable vibration without an exact preidentified process model.
- Identified feed-drive dynamics and nonlinear friction from experiments, developed **MATLAB** and **Simulink** models, and used **dSPACE** to evaluate motion and controller performance.

Research Assistant | Feb. 2022 - Sep. 2023

Smart Electromechanical Energy Conversion Systems Laboratory

- Identified planar voice-coil actuator dynamics from chirp-response measurements in **MATLAB**, achieving **90.75% model fit** and using the model to design a positioning controller.
- Designed a digital positioning controller from the **identified model and implemented** it in **C** on an **STM32** microcontroller; the response to a 10 mm step settled in under 0.5 seconds with no observed overshoot and near-zero steady-state error.
- Integrated magnetic position sensing, servo-amplifier circuitry, and STM32 control into a 1 kHz platform for actuator testing and closed-loop experiments.
- Characterized friction introduced by a syringe-pump load and compared a nonlinear **Simulink** model with measured positioning responses.

Research Intern | Jul. 2021 - Jan. 2022

Bio-MEMS and LOC Lab

- Developed a **C library for an ATmega8** microcontroller to execute measurement routines for a coin-sized potentiostat.
- Designed and analyzed low-pass filtering circuitry and implemented Arduino measurement routines to support potentiostat development.

Selected Engineering Projects

Precision Motion Control for Musical Record Manufacturing | 2025

ASPE Student Challenge | Oregon State University | **First-Place Team Award** | KLA and ASML among challenge sponsors

- Characterized a voice-coil-driven flexure stage at five positions using sine-sweep testing and applied gain-scheduled inverse filtering to compensate for position-dependent dynamics.
- Designed a feedback controller and position-dependent inverse filters, achieving approximately **1 Hz tracking bandwidth with nominal gain and phase margins of 17 dB and 83°**, respectively.
- Implemented real-time motion control in **LabVIEW** on **NI myRIO** and collaborated on hardware integration, alignment, and experimental testing, contributing to the team's first-place finish.

MATLAB GUI for Beam Buckling Analysis

Mechanics of Materials II Course Project | 2020

- Co-developed a MATLAB GUI with a course teammate to calculate beam load limits across selectable materials, cross sections, support conditions, and axial or eccentric loading, with custom inputs and unit conversion.

ArmPi Robotic Writing - Trajectory Generation and Experimental Integration

ROB 515 Course Project | Oregon State University | 2026

- Generated letter trajectories in **MATLAB**, discretized each path into 20 pen-up/pen-down waypoints, and implemented the motion sequence in **Python** on a **Raspberry Pi-controlled robot**.
- Integrated and calibrated the physical writing setup by stabilizing the marker and developing an iterative Python height-test routine for reliable paper contact.
- Troubleshoot vibration and path-execution errors by reducing waypoint density, increasing X-Y motion time to 1.8 seconds, and relocating the writing region closer to the robot base to reduce observed vibration during writing.

PiCar-X Autonomous Mobile Robot - Perception, Control, and Systems Integration

ROB 515 Course Project | Oregon State University | 2026

- Implemented and tested Python modules for camera and grayscale sensing, ultrasonic ranging, **steering-servo control, and DC-motor actuation**, including obstacle response at approximately 15 cm.
- Organized sensing, interpretation, and control using RossROS producer-consumer processes and inspected intermediate image, bus, and command outputs to isolate faults across the perception-to-actuation workflow.

Mechanical Robot Gripper Redesign

ME 512 Course Team Project | Oregon State University | 2025

- Redesigned and simulated a robot-gripper linkage in **Siemens NX**, evaluating end-effector kinematics, force transmission, mechanical advantage, and the effect of a two-stage gear train.

Technical Skills

Programming: MATLAB, Simulink, LabVIEW, Python, C, C++

Dynamics and Control: System identification, dynamic modeling of mechanical systems, signal processing, frequency-response analysis, feedback and adaptive feedforward control, vibration and stability analysis

Embedded Systems and Testing: STM32, NI myRIO, dSPACE, Arduino, Raspberry Pi; sensor and actuator integration, data acquisition, calibration, hardware testing and validation

Design and Electronics: SolidWorks, Siemens NX, COMSOL Multiphysics, NI Multisim; circuit and schematic analysis

Selected Publications

- Bahtiyar, K., Bozcu, L., **Shantiaeezade, Seyed Mahmood**, Sencer, B., Ikeda, R., & Suzuki, N. "ADAPTIVE TOOL ECCENTRICITY COMPENSATION USING MACHINE TOOL FEED DRIVES." MM Science Journal (2025).
- Kawana, Y., **Shantiaeezade, Seyed Mahmood**, Sencer, B., Ikeda, R., & Suzuki, N., "Low frequency feed modulation assisted milling for chatter avoidance". CIRP Annals (2025).
- **Shantiaee Zade, Seyed Mahmood**, Robati, M., Nejad, M. G., Maleki, S., & Sadighi, A. (2022, November). "Optimal design and control of a planar voice-coil actuator for micropositioning applications". In 2022 10th RSI International Conference on Robotics and Mechatronics (ICRoM) (pp. 439-444). IEEE.

Teaching Experience

Teaching Assistant | Oregon State University | 2024 - 2025

Courses: Computer-Aided Design and Manufacturing; Computer Control of Manufacturing Processes; Lean Manufacturing Systems Engineering; Composite Manufacturing

Teaching Assistant | University of Tehran | 2021 - 2022

Courses: Automatic Control; Design of Machine Elements I